

CY 600 : Research Methodology for Chemistry

Course Code : CY 600

Course Name : Research Methodology for Chemistry

L-T-P-C : 1-0-0-1

Intended for : PhD

Prerequisite :

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Literature Survey:** Literature search via journals and using various search engines. Knowledge of journal abbreviations and parameters, and different publishing platforms. Citation matrices. (3 Hours)
- **Scientific Writing, Communication, and Ethics:** Structure of research paper/article/literature review. Scientific writing and presentation techniques. Ethical conduct in scientific research and knowledge of plagiarism detecting tools. Bioethics in experiments with living organisms. (3 Hours)
- **Chemical Safety:** Safety protocols in the laboratory, first aid, and emergency response procedures. Proper disposal methods for waste chemicals. (2 Hours)
- **Data Analysis and Chemistry Software:** Quantitative and Qualitative data analysis. Knowledge of data analysis software, data reliability, and validity. Exposure to various chemistry software. (5 Hours)
- **Intellectual Property:** Briefing about intellectual property, patents, copyrights, and trademarks. (1 Hour)

Textbooks:

1. NA

References:

1. Fraenkel F. J. & Warren N. E., **How to Design and Evaluate Research in Education**, 4th Edition, McGraw-Hill, 2002.
2. Dean, J. R., Jones, A. M., Holmes, D., Reed, R., Weyers, J. & Jones, A., **Practical skills in chemistry**, 2nd Edition, Prentice-Hall, 2011.
3. Hibbert, D. B. & Gooding, J. J., **Data analysis for chemistry**, Oxford University Press, 2006.
4. Levie, R. de., **How to use Excel in analytical chemistry and in general scientific data analysis**, Cambridge University Press, 2001.
5. **Chemical safety matters – IUPAC – IPCS**, Cambridge University Press, 1992.
6. Internet resources.